

GAWorld-Bench: A Layered Validation Framework for LLM-Based Artificial Societies

Anonymous submission

Abstract

Large language models make it easy to construct artificial societies whose agents remember, converse, and react to interventions. It remains difficult to tell when such a system supports scientific inference rather than plausible storytelling. We introduce GAWorld-Bench, a claim-centered validation framework for LLM-based artificial societies. It separates four scientific claim layers—macro empirical fit, emergent dynamics, counterfactual validity, and individual consistency—from a cross-cutting reproducibility/cost gate, and requires every result to carry an evidence-provenance label. Coverage is reported beside, rather than absorbed into, layer outcomes. We operationalize provenance, macro and counterfactual checks, and a deterministic-mode gate; the remaining tests are specified but unimplemented. We apply this audit retrospectively to GAWorld, an urban multiagent simulator. It reveals three failure modes that conventional headline scores obscure: parameter recovery can be circular; two aggregation windows yielded traffic contrasts differing by roughly 49-fold; and archived outputs from incompletely recorded repository states are not comparable enough for causal inference. A repository headline scorecard is also numerically identical to synthetic fixture output, while its lineage is not preserved. We do not conclude that GAWorld predicts real populations. Instead, the case study shows how provenance-aware, layered evaluation makes missing evidence visible and constrains the claims an artificial-society experiment can support.

Introduction

Generative agents combine language models with memory, planning, and social interaction to produce open-ended behavior (Park et al. 2023). Recent systems extend this idea toward simulations of individuals grounded in interviews and surveys (Park et al. 2024), while agent benchmarks evaluate interactive task completion (Liu et al. 2024; Ma et al. 2024) and social competence (Zhou et al. 2024). These advances create a tempting inference: if agents act plausibly, a society of such agents can be used to study social mechanisms or policy counterfactuals.

That inference is not generally warranted. A coherent diary does not validate a population statistic; matching a popula-

tion statistic does not establish an emergent mechanism; and a directional response to one intervention does not establish a stable counterfactual instrument. Moreover, a simulator can pass a benchmark by recovering quantities encoded in its own parameters, or a benchmark implementation can pass on synthetic fixtures even when no empirical model run passes. Work on agent-based models (ABMs) has long distinguished verification, calibration, and validation (Sargent 2013; Fagiolo, Moneta, and Windrum 2007), but LLM-society reports often combine heterogeneous evidence into a realism score or a small set of demonstrations.

We ask: *when does an LLM-based artificial society qualify as a scientific instrument rather than a system that generates plausible narratives?* Our answer is operational rather than philosophical. Every claim must be paired with a validation layer, admissible evidence, provenance, and an explicit outcome: pass, fail, or unassessed. Reproducibility is a gate, not another score that favorable dimensions can offset.

This paper contributes:

- a layered framework separating four scientific claim types from a reproducibility/cost gate;
- GAWorld-Bench, which operationalizes provenance, macro and counterfactual checks, coverage reporting, and a deterministic-mode gate while specifying the remaining tests; and
- a retrospective audit showing how window choice, non-comparable archives, missing tracks, and fixture-like scorecards can produce overconfident conclusions if evidence classes are collapsed.

Our contribution is not a claim that GAWorld is already a valid model of a real city. The audit is a worked example of how an evaluation framework can reject unsupported conclusions and specify the evidence still required.

Related Work

Generative societies and synthetic respondents. Generative Agents demonstrated memory, reflection, planning, and social diffusion in a small interactive town (Park et al. 2023). Later work grounded agents in self-reports from 1,052 participants and evaluated held-out survey and behavioral outcomes (Park et al. 2024). Language models have also been used to approximate survey samples (Argyle et al. 2023). This line of work motivates richer artificial societies, but it also

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motivates careful limits: generated respondents need not reproduce the experiences of real identity groups (Dillion et al. 2023; Wang, Morgenstern, and Dickerson 2025). GAWorld-Bench treats external validity as an empirical question rather than a consequence of fluent behavior.

Evaluation of LLM agents. AgentBench evaluates reasoning and decision making across interactive environments (Liu et al. 2024); AgentBoard adds process-oriented progress measures (Ma et al. 2024); and SOTOPIA evaluates social goal achievement and interaction quality (Zhou et al. 2024). These benchmarks evaluate important capabilities of an agent. Our target is different: a society makes claims at several levels—individual, aggregate, dynamical, and counterfactual—whose evidence cannot be interchanged. GAWorld-Bench therefore evaluates the mapping from claim to evidence, not only task performance.

Validation of simulation models. ABM research emphasizes transparent model description through protocols such as ODD (Grimm et al. 2006), empirical validation against multiple kinds of observations (Windrum, Fagiolo, and Moneta 2007; Fagiolo, Moneta, and Windrum 2007), and separation of verification from validation (Sargent 2013). Generative social science further stresses explaining macro patterns from micro rules (Epstein 2006). We adapt these principles to stochastic LLM agents and add two practical safeguards: explicit evidence provenance and a trust gate. Placebo interventions follow the logic of negative controls, which expose residual bias when an effect should be absent (Lipsitch, Tchetgen, and Cohen 2010).

A Layered Validation Framework

Figure 1 organizes validation by the proposition being tested. The ordering is not a claim that one layer implies another. It is a discipline for preventing evidence migration: results admissible for one proposition cannot silently validate another.

L1: macro empirical fit. This layer asks whether simulated aggregates fall near external observations. For simulated value x_i , anchor a_i , and relative tolerance τ_i , a bounded diagnostic is

$$s_i = \max\left(0, 1 - \frac{|x_i - a_i|}{|a_i|\tau_i}\right). \quad (1)$$

The vector $(s_1, \dots, s_k)$ should remain visible even if a mean is reported. Most importantly, every metric is tagged as *input-derived* or *emergent*. Recovering a tax schedule, consumption curve, or demographic distribution directly encoded in initialization is a software or calibration check, not independent validation.

L2: emergent dynamics. This layer asks whether unprogrammed temporal or network regularities appear: diffusion shapes, homophily, community formation, long-tailed wealth, or other domain-specific stylized facts. A pass requires a preregistered statistic, tolerance, time horizon, and replication rule. A qualitative anecdote or a pattern inferred from an initialization snapshot is insufficient.

L3: counterfactual validity. For event e , metric m , matched baseline B , event run E , and endpoint T , define

$$\Delta_{e,m}^{\text{final}} = y_{m,T}^E - y_{m,T}^B. \quad (2)$$

A known-sign test checks whether $\text{sign}(\Delta_{e,m}^{\text{final}})$ matches a direction fixed before examining output. The endpoint is a diagnostic, not a universal estimator; an event-time average or trajectory model may be preferable when the estimand demands it. GAWorld-Bench therefore requires the aggregation window to be declared. A placebo event additionally requires $|\Delta_{e,m}| < \epsilon$ for unaffected metrics. Replicated runs must report uncertainty; a single seed can provide a case, not a significance claim.

L4: individual consistency. This layer tests whether an agent’s decisions remain grounded in its profile and recorded experience. Possible probes include traceable recall, persona agreement, cross-day contradiction rate, and behavior under memory ablations. Fluency is not a metric. LLM judges, if used, need calibration, judge-model sensitivity checks, and access to ground-truth traces.

Gate: reproducibility and cost. The cross-cutting gate measures reproducibility under a protocol appropriate to the simulator, together with parse or tool failure rate, tokens, and wall-clock cost. A deterministic execution mode can require same-seed state equality; a stochastic mode instead requires replicated distributional agreement and uncertainty estimates. Failure of the *declared* protocol blocks claims that depend on paired runs, not every possible stochastic simulation. An untested gate yields UNVERIFIED; a failed declared protocol yields UNTRUSTWORTHY. Neither state can be averaged away by favorable scientific layers.

Three-valued outcomes and coverage. Each configured scientific test returns pass, fail, or unassessed; the gate separately returns OK, UNVERIFIED, or UNTRUSTWORTHY. Let n_{eval} be evaluated tests and n_{cfg} configured tests. Coverage $c = n_{\text{eval}}/n_{\text{cfg}}$ is reported beside any within-layer score but is not multiplied into it: doing so would be algebraically equivalent to imputing missing tests as zero. Missing evidence remains N/A rather than success or failure.

GAWorld-Bench

GAWorld-Bench instantiates the framework as a filesystem audit over state histories, economy ledgers, social traces, intervention records, and matched comparison outputs. It does not require one monolithic dataset. Instead, it assigns every artifact one of four provenance labels before scoring: REAL, DIAGNOSTIC, SYNTHETIC, or INCOMPLETE.

Figure 2 and Table 1 make provenance a first-class part of the result. This matters because synthetic data can legitimately verify a scoring path while saying nothing about the simulator. GAWorld-Bench stores layer scores separately and reports coverage. The audited implementation also emits a legacy coverage-discounted composite; we retain it only as a non-decisional software output and do not interpret it because it conflates evidence quality with missingness.

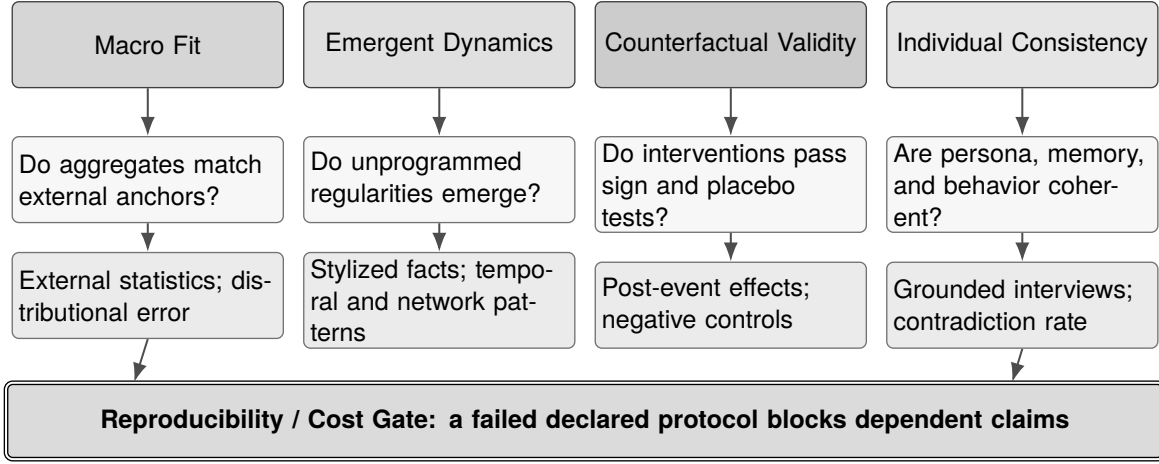


Figure 1: The framework separates four scientific claim layers and a cross-cutting engineering gate. Each scientific layer receives pass, fail, or unassessed status. A failed declared reproducibility protocol blocks dependent claims; an untested gate leaves them unverified.

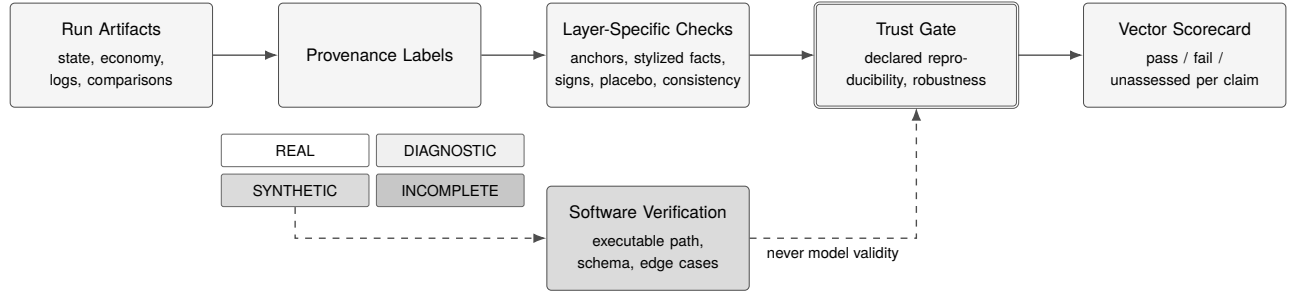


Figure 2: Provenance is assigned before layer-specific checks. Synthetic fixtures are routed to software verification, not model validity. Incomplete artifacts remain N/A. Only admissible evidence reaches the trust gate and vector scorecard.

Table 1: Evidence classes and permitted claims.

Class	Meaning	Permitted use
Real	Existing simulator run	Descriptive evidence; inference only with adequate replication
Diagnostic	Re-analysis of real output	Measurement sensitivity and failure analysis
Synthetic	Fabricated structural fixture	Harness execution, schemas, and edge cases only
Incomplete	Missing or insufficient output	Report N/A; exclude from aggregates

Counterfactual track. The implemented causal track combines known-sign interventions, placebo tests, and a deterministic-mode subtest. The emitted report retains the components, gate state, and coverage; its legacy weighted scalar is never sufficient to claim validity. In multi-seed mode,

a known-sign item counts as supported only if its confidence interval excludes zero and has the preregistered sign.

Artifact auditability. The harness reads long-format state histories and comparison CSVs, records which directories are missing required outputs, and archives human-readable reports. No LLM is needed to re-score completed artifacts. This separation is important: model execution may be expensive and stochastic, while validation logic should remain inspectable and deterministic.

Auditing GAWorld

Audit scope. We retrospectively inspected selected GAWorld artifacts and benchmark reports; we did not re-run the simulator. The selection covers artifacts used by the benchmark’s configured tracks plus known incomplete emotion, network, and memory reports, but it is not a repository-wide exhaustive inventory. Because the working tree and archived run metadata do not identify a common frozen release, the audit evaluates admissible claim strength, not predictive accuracy or stochastic stability. Table 2 separates scientific outcomes from the gate.

Table 2: GAWorld evidence state under the layered audit.

Layer	Outcome	Reason
Macro fit	Unassessed	Two one-agent snapshots do not estimate a population distribution
Emergent dynamics	Unassessed	Emotion state files absent; network report contains initialization only
Counterfactual	Unassessed	Single observations; archive comparability and uncertainty unavailable
Individual consistency	Unassessed	Only one memory treatment completed both phases
Repro. / cost	Unassessed	Cross-seed robustness, failure rate, and cost incomplete; gate UNVERIFIED

Macro fit is neither stable nor independent. The current economy snapshot contains one agent with Engel coefficient 0.48 and savings rate 0.05, against benchmark-configured targets 0.288 and 0.35. An earlier real report contains another one-agent snapshot with values 0.30 and 0.25. The urban Engel anchor is consistent with an official Chinese statistic (National Bureau of Statistics of China 2025); the savings target lacks a matched external definition and is only an internal benchmark setting. Neither one-agent output estimates a population distribution. Furthermore, the simulator directly parameterizes parts of consumption and saving. We therefore classify these comparisons as implementation diagnostics, not external validation. A single recorded conservation-audit row has zero monetary drift; this checks an invariant for that row only.

Aggregation windows change the conclusion. Table 3 reproduces existing audit reports. For a traffic restriction, the whole-run mean difference in mobility intent was $+0.0068$, while the recorded endpoint difference was $+0.3368$ —about 49 times larger. For a tax cut, economic security changed sign: -0.0086 for the whole-run mean versus $+0.0201$ at the endpoint. These are not estimates of real policy effects. They demonstrate that an unstated aggregation window can suppress or reverse a benchmark conclusion. An endpoint can itself be noisy; the remedy is to declare the estimand and report trajectories or post-event windows, not to select whichever statistic passes.

Non-comparable archives block causal claims. A June 20 report gave layoff endpoint differences of -0.0498 for economic security and $+0.1717$ for stress. A later batch contained one observation per intervention and gave $+0.0826$ and $+0.0078$, respectively. The latter batch cannot produce a confidence interval despite being processed by the multi-seed path. Code, prompts, population, provider, and matched-baseline metadata are not jointly preserved across these reports. We therefore do not attribute the disagreement

to stochastic instability; we treat it as evidence that comparability, replication, version pinning, and uncertainty are required.

Synthetic success is not model success. The repository’s current headline scorecard reports macro values near 0.29 and 0.328, four exact known-sign effects ($+0.08$, -0.12 , $+0.09$, $+0.06$), a perfect placebo, and perfect determinism. Those values match the benchmark’s synthetic-fixture generator exactly. This numeric fingerprint is strong evidence of fixture-like output but, without a preserved generation command or input manifest, does not by itself prove lineage. The scorecard is therefore admissible only as a software-path check and is excluded from every claim about GAWorld. This illustrates why provenance must be recorded at creation.

What the audit establishes. The case study establishes that GAWorld-Bench can expose circular checks, measurement sensitivity, non-comparable evidence, fixture-like fingerprints, and unassessed claims. It does not establish population realism, emergent social laws, or policy-effect accuracy. This distinction is the intended output of a validation instrument.

Limitations, Ethics, and Reproducibility

Limitations. GAWorld is one case study, and several proposed tracks are not yet implemented. The audit is retrospective and spans changing code and configurations; this makes it useful for provenance diagnosis but unsuitable for comparing model versions. Known-sign tests are coarse and can reward hard-coded responses unless paired with mechanism ablations and unseen interventions. Endpoint effects can be unstable, and placebo thresholds require domain justification. The framework still needs evaluation across other artificial-society platforms and calibration of its decision rules.

Ethical statement. Artificial societies can help debug hypotheses before costly field work, but they can also launder model bias into apparently quantitative policy advice. Synthetic agents do not consent, experience harm, or represent affected communities. Their outputs should not replace empirical data or stakeholder participation, especially for high-impact decisions. Provenance labels and unassessed states reduce, but do not eliminate, this risk. We therefore disclose negative and missing results and avoid claims about the modeled policies themselves.

Reproducibility statement. The audit maps each reported number to an archived CSV, JSON, or Markdown report. The package records artifact paths and hashes, aggregation definitions, and the distinction between source artifacts and derived diagnostics. Its offline commands verify selected extractions; they do not regenerate a complete scorecard from a frozen run because the necessary lineage is absent. A supplement provides formulas, the evidence ledger, and verification commands. Full reproduction of new simulations would additionally require model/provider versions, prompts, seeds, and API behavior, which the available artifacts do not completely preserve; the corresponding gate remains unverified.

Table 3: Selected diagnostic values from existing real reports. Each row is unreplicated; no confidence interval or policy-effect claim is warranted. “Whole” is the recorded whole-run mean difference and “final” is the recorded endpoint difference.

Report	Intervention / metric	Δ_{whole}	Δ_{final}	n	Expected	Audit reading
2026-06-15	Traffic / mobility intent	+0.0068	+0.3368	1	+	49-fold window sensitivity
2026-06-15	Tax cut / economic security	-0.0086	+0.0201	1	+	Sign changes by window
2026-06-20	Layoff / economic security	-0.0319	-0.0498	1	-	Expected sign in one run
2026-06-20	Layoff / stress	+0.1446	+0.1717	1	+	Expected sign in one run
2026-07-09	Layoff / economic security	—	+0.0826	1	-	Opposite sign in later batch
2026-07-09	Layoff / stress	—	+0.0078	1	+	Different magnitude; insufficient

Conclusion

LLM-based artificial societies need validation organized around claims, not plausibility. GAWorld-Bench separates evidence layers, records coverage, preserves provenance, and gates paired scientific claims on a declared reproducibility protocol. Its GAWorld audit shows that these safeguards change what can responsibly be concluded: several attractive headline results are diagnostics or fixture-like, while the empirical tracks remain incomplete or non-comparable. Making that boundary explicit is progress. Future work should complete multi-seed, placebo, mechanism-ablation, persona-grounding, and cost evaluations across multiple platforms.

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